

Mask Out: A Novel Regularization Method for Code-Switching Language Identification

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Abstract

Code-switching, the practice of alternating between languages, presents significant challenges for natural language processing. This paper introduces *Mask Out*, a novel method for word-level code-switching language identification that leverages multilingual BERT (mBERT) and a token-masking regularization technique. By selectively obscuring language-specific cues during training, *Mask Out* enables the model to focus on underlying code-switching patterns, enhancing its ability to generalize across diverse linguistic contexts.

We evaluate our approach on the SPA-ENG (Spanish-English) dataset for training and test its generalization on NEP-ENG (Nepali-English), an unseen language pair. Experimental results demonstrate that the optimal masking probability of 0.1 yields the best performance, as measured by F1-score. Higher masking probabilities, exceeding 0.3, were found to obscure critical information, leading to performance degradation. These findings highlight the importance of balancing regularization strength to achieve robust and effective learning.

Our results demonstrate the potential of *Mask Out* as a robust regularization method for code-switching tasks, offering a promising direction for future research in multilingual NLP.

The implementation of *Mask Out* is publicly available at: https://github.com/Sean652039/code_switching_LID.

1 Introduction

Code-switching is a linguistic phenomenon in which multilingual individuals switch between languages within the same conversation, (Poplack, 1980). The number of multilingual speakers worldwide, i.e., those who speak multiple languages, significantly exceeds the number of monolingual speakers, highlighting the importance of developing NLP technology capable of handling code-switched data. Despite this, research on code-

switching related tasks only gained significant attention in recent years, (Sitaram et al., 2019), driven by two main factors: (1) The need to analyze social media content. Previously, code-switching was more common in spoken language, but with the rise of social media, multilingual users increasingly mix languages in written form; (2) The growing availability of voice-operated devices. As smart assistants become more widespread, it has become clear that assuming users will communicate with NLP systems as monolinguals is limiting and does not address the needs of real-world users, (Bawa et al., 2020). To address the challenges posed by code-switching, it is crucial to understand not just the occurrence of multiple languages but the specific points where language switching happens within a conversation.

Code-switching tasks in NLP are typically categorized into detection and prediction. Detection identifies language boundaries within a code-switched utterance and determines the language of words or segments around these boundaries (Wang et al., 2019). In contrast, prediction forecasts switching points by utilizing models trained on bilingual corpora to capture contextual cues (Solorio and Liu, 2008). Both tasks play a crucial role in understanding and modeling code-switching behavior for real-world applications.

2 Related Work

Studies on code-switching have investigated the factors and patterns that influence its occurrence, examining its manifestations across diverse contexts and disciplines. Prior research highlights several key drivers of code-switching behavior:

1. **Social and Contextual Factors:** Code-switching often arises in response to social and situational contexts. Early research by (Poplack, 1980) and later by (Myers-Scotton, 1993) indicated that the choice of language can be influenced by fac-

tors such as the audience, the topic of conversation, and the setting. Speakers may switch languages to accommodate the preferences of their interlocutors or to signal group membership and cultural identity. For instance, switching to a heritage language during a conversation can serve as an act of solidarity or cultural expression, reinforcing social bonds within a group. This social dimension makes code-switching a dynamic and context-dependent phenomenon, driven by the need to navigate social relationships effectively.

2. **Grammatical Constraint and Syntactic Structure:** Research has identified certain syntactic positions where code-switching is more likely to occur. (Poplack, 1980) proposed the equivalence constraint, suggesting that switches tend to happen where the grammatical structures of both languages are congruent, such as between noun phrases or at clause boundaries. (Sankoff and Poplack, 1981) further emphasized that maintaining grammatical integrity across languages is crucial in code-switched speech. This aligns with the observation that bilingual speakers often switch languages at points in the sentence where the structures of both languages permit it, ensuring that the switch does not violate the grammar of either language.

3. **Lexical and Vocabulary-Related Influences:** Lexical gaps and vocabulary-related factors play a significant role in code-switching. (Gumperz, 1982) noted that speakers often switch languages when a concept is more naturally or accurately conveyed in the other language, especially when the vocabulary in one language lacks a direct equivalent. This is commonly observed in cases involving borrowed words, technical terms, or cultural-specific references. Such lexical considerations drive speakers to code-switch to better express their intended meaning or to ensure that their message is understood more precisely.

4. **Emotional and Pragmatic Functions:** Emotional and pragmatic motivations also influence code-switching behavior. (Dewaele, 2010) found that bilinguals may switch languages to convey emotions more authentically or to achieve specific pragmatic effects, such as humor or emphasis. For example, speakers might switch to a language in which they feel more emotionally connected to express strong feelings or to soften the tone of the conversation. Code-switching, in this sense, can serve as a tool for adding nuance to communication and shaping the listener’s interpretation.

5. **Impact of Media and Digital Communication:**

The rise of digital communication and social media has significantly impacted code-switching patterns. (Androutsopoulos, 2013) observed that code-switching in written form, which was less common historically, has become more prevalent with the growth of online platforms. In social media posts, language switches often occur when users address different audiences, tag others, or quote content from different sources. This online environment encourages fluid language use, with speakers frequently alternating between languages to engage with a diverse audience or convey messages more effectively.

6. **Structural and Statistical Cues:** In recent years, efforts have been made to predict code-switch points based on linguistic and statistical features. (Solorio and Liu, 2008) explored the use of structural cues, such as word frequency and part-of-speech tags, to model code-switching behavior. Additionally, features like entropy have been employed to capture uncertainty in language choice, which can indicate a higher likelihood of a switch. These predictive approaches aim to uncover patterns in language usage that could help improve the performance of NLP systems in handling code-switched text.

From a computational perspective, various methods have been developed to address code-switching tasks, including detection and language identification. Early approaches utilized sequence-based models like Long Short-Term Memory (LSTM) networks, which achieved strong performance in multilingual scenarios (Samih et al., 2016). Similarly, Recurrent Neural Networks (RNN) augmented with part-of-speech (POS) features effectively identified switching points by factorizing language-specific outputs (Adel et al., 2013). More recently, transformer-based models have set new benchmarks for code-switching tasks. For instance, (Zhou et al., 2020) introduced a dual symmetric encoder to enhance acoustic representations of code-switched text. Furthermore, unsupervised methods like Hidden Markov Models (HMM) have addressed the challenges of limited labeled data, offering practical solutions for multilingual NLP (Rijhwani et al., 2017).

3 Experimental Framework

3.1 Dataset

For this project, we leverage two datasets from the Linguistic Code-switching Evaluation (LinCE)

benchmark (Aguilar et al., 2020), which provides ten corpora across four code-switched language pairs and four NLP tasks. Specifically, we focus on the Spanish-English (SPA-ENG) and Nepali-English (NEP-ENG) datasets.

We hypothesize that code-switching patterns may exhibit commonalities across language pairs that share a common language—in this case, English. To test this hypothesis, we train our model on the SPA-ENG dataset using the proposed token-masking regularization technique and evaluate its performance on the NEP-ENG dataset. This cross-dataset evaluation aims to measure the model’s generalization capacity and its ability to recognize and adapt to code-switching patterns in a different language pair and linguistic context.

3.2 Pre-processing

The original datasets contain code-switched text annotated with eight labels, representing language pairs, usernames, emoticons, and other categories. To focus on the impact of our proposed regularization method, we adopt the evaluation approach outlined by (Molina et al., 2019). Specifically, we simplify the label space by retaining only the lexicons for Lang1 and Lang2, merging all other labels into a single category labeled as "other."

For testing datasets, where English is consistently labeled as Lang1, we ensure consistency by applying the same labeling scheme to the training datasets, even though these testing datasets remain unseen during training.

To address label imbalance, we calculate class weights systematically by analyzing the label frequencies in the training dataset. After quantifying class distributions, we apply a square root transformation to the frequencies, followed by normalization, to produce a balanced weight tensor. This weighting strategy helps mitigate bias and improve model performance for minority classes in the classification task. Figure 1 provides a visual representation of the label distribution in the training and test datasets.

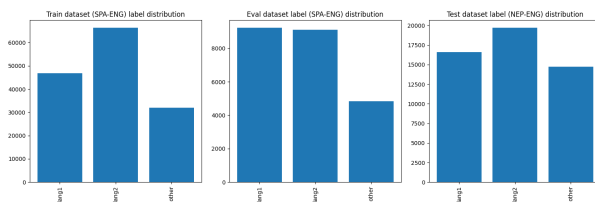


Figure 1: Label distribution in the dataset.

3.3 Evaluation Metrics

We evaluate the model’s performance using several metrics, including precision, recall, F1-score (macro and weighted), and accuracy. To further assess the model’s generalization capacity, we evaluate its performance on an unseen dataset, examining whether its predictions accurately capture language-switching tendencies across diverse contexts, as outlined in Section 3.1.

The experimental results provide insights into the effectiveness of the proposed method in detecting code-switching points, highlighting the strengths and limitations of the approach.

3.4 Transformer-Based Models

As discussed in Section 2, code-switching often follows specific linguistic patterns within bilingual contexts. To effectively capture these patterns, we leverage the Transformer architecture, known for its ability to model long-range dependencies in text. Specifically, we fine-tune Multilingual BERT (mBERT), a pre-trained model optimized for multiple languages, which makes it well-suited for handling code-switched text.

By fine-tuning mBERT, we adapt the model to predict code-switch probabilities at the word level. This approach harnesses the model’s robust multilingual capabilities to address the inherent complexities of code-switching, enabling it to better identify and generalize patterns across diverse language pairs.

3.5 Early Stopping

To prevent overfitting during training, we employ early stopping as a regularization technique. This approach monitors the validation loss and terminates training if no improvement is observed over a predefined number of epochs. By halting training at an optimal point, early stopping minimizes the risk of overfitting to the training data, thereby improving the model’s ability to generalize to unseen data.

3.6 Mask Out

The regularization technique proposed in this study draws inspiration from Dropout, a widely used method to mitigate overfitting in neural networks (Nitish, 2014), and MaskLID, a recent framework for sentence-level code-switching language identification (Kargaran et al., 2024). Building on these approaches, we introduce a novel token-masking

regularization method to improve word-level language identification in code-switched data.

Our method leverages the pretraining objective of BERT, which involves predicting masked tokens, to enhance its effectiveness for this task. We hypothesize that combining token masking with this pretraining objective can produce synergistic benefits, enabling the model to better capture linguistic patterns in code-switched text.

The core mechanism of the proposed method involves probabilistic token masking during the training phase. By randomly obscuring tokens with a predefined probability p , the model is encouraged to rely on broader contextual cues rather than surface-level linguistic markers. This promotes a more robust and context-aware understanding of language-switching patterns.

To clarify the implementation, Figure 2 provides an example from the SPA-ENG training dataset (sample 28), illustrating the token-masking process. This example demonstrates how selective token obscuration compels the model to adopt more sophisticated strategies for language identification.

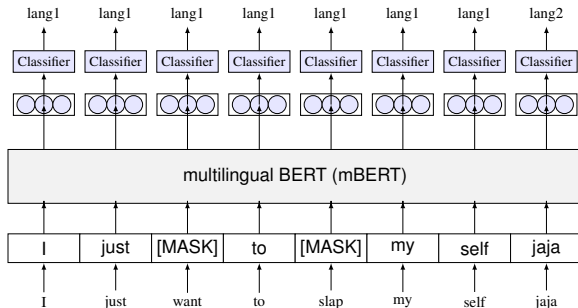


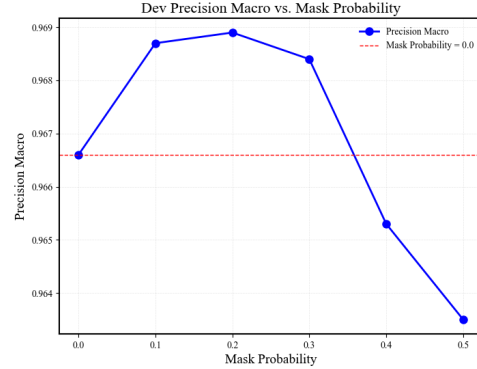
Figure 2: Visualization of mBERT’s implementation (sentence number 99, from the (SPA-ENG) training dataset).

4 Results and Discussion

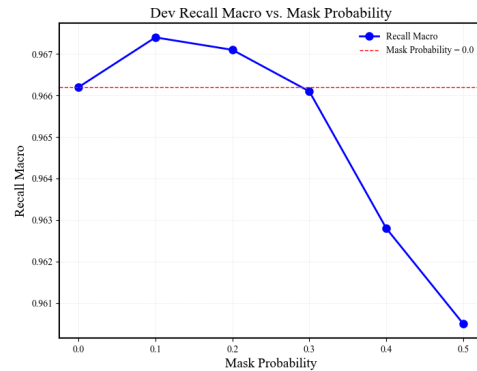
To assess the effectiveness of the proposed token-masking regularization method, we perform experiments on the LinCE benchmark datasets, focusing on the SPA-ENG and NEP-ENG language pairs. The evaluation involves comparing models trained with varying mask-out probabilities, ranging from 0.0 to 0.5.

Figure 3 visualizes the impact of mask-out probability on key performance metrics, including precision, recall, and accuracy. The results reveal distinct trends, demonstrating how the choice of mask-out probability influences the model’s ability to generalize across different linguistic contexts.

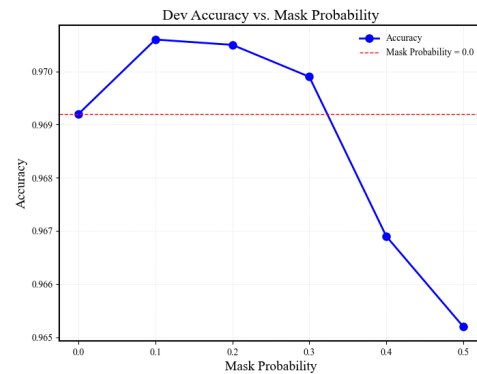
These findings underscore the importance of tuning the regularization strength to achieve an optimal balance between model robustness and performance.



(a) Precision Macro vs. Mask Probability



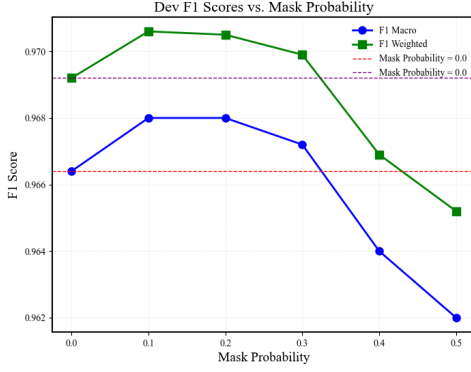
(b) Recall Macro vs. Mask Probability



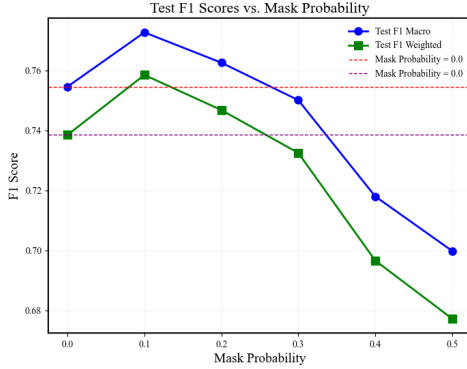
(c) Accuracy vs. Mask Probability

Figure 3: Performance metrics with different mask-out probabilities.

Figure 4 depicts the F1 scores across different mask-out probabilities. Specifically, we evaluate both the macro and weighted F1 scores on the development and test datasets, providing a comprehensive view of the model’s performance under varying regularization strengths.



(a) F1 score on the dev dataset



(b) F1 score on the test dataset

Figure 4: F1 scores with different mask-out probability. The overall trend suggests that a mask-out probability of 0.1 yields the best performance.

Table 1 summarizes the results of these experiments, emphasizing the impact of our regularization method on key performance metrics. The findings indicate that a mask-out probability of 0.1 achieves the best performance across all metrics. This supports our hypothesis that selectively obscuring language-specific cues during training enhances the model’s capacity to capture code-switching patterns more effectively.

5 Analysis

Understanding how the model distributes its logits across classes provides valuable insights into its decision-making process and the impact of regularization. To analyze this, we visualize the logits produced by models trained with varying mask-out probabilities. Figures 5 and 6 present 3D and 2D projections of the logits, respectively, offering complementary perspectives on class separability and clustering.

The 3D visualizations reveal how the logits span across three selected dimensions, illustrating the

relationships between classes (lang1, lang2, and other). These plots capture intricate patterns in the decision space, such as the extent of overlap or separation between classes. This higher-dimensional view is particularly effective at identifying subtle interdependencies and variations in the model’s confidence across different mask-out probabilities.

On the other hand, the 2D projections condense the information into two dimensions, simplifying the representation while preserving key insights. These visualizations make it easier to observe class boundaries, cluster formation, and potential areas of confusion between lang1, lang2, and other. They also provide a clearer depiction of how the mask-out probability influences the distribution of logits and, consequently, the model’s behavior.

At lower mask-out probabilities (e.g., 0.0 and 0.1), the logits form well-defined clusters, indicating that the model confidently distinguishes between classes. However, as the probability increases (e.g., beyond 0.3), the clusters become more diffuse, reflecting reduced confidence and a growing reliance on noisy or insufficient input. These patterns align with our experimental results, which show that an optimal mask-out probability (0.1) balances regularization and information retention, while higher probabilities compromise performance.

6 Conclusion

This project introduces *Mask Out*, a novel regularization method for code-switching language identification that leverages multilingual BERT (mBERT) and a token-masking technique. By obscuring language-specific cues during training, *Mask Out* encourages the model to focus on underlying code-switching patterns, enhancing its ability to capture language-switching tendencies effectively.

Experimental results on the SPA-ENG and NEP-ENG datasets demonstrate that an appropriate mask-out probability significantly improves model performance, with 0.1 yielding the best results across key metrics. These findings underscore the potential of *Mask Out* as a robust regularization method for code-switching tasks and highlight its promise for advancing multilingual NLP research.

7 Future Work

We hypothesize that the proposed regularization method could benefit a range of other NLP tasks,

Mask Out Probability	Spanish-English (SPA-ENG)						Nepali-English (NEP-ENG)	
	Dev Loss	F1 Macro	F1 Weighted	Precision Macro	Recall Macro	Accuracy	F1 Macro	F1 Weighted
0.0	0.0268	0.9664	0.9692	0.9666	0.9662	0.9692	0.7546	0.7385
0.1	0.0253	0.9680	0.9706	0.9687	0.9674	0.9706	0.7727	0.7585
0.2	0.0257	0.9680	0.9705	0.9689	0.9671	0.9705	0.7626	0.7468
0.3	0.0265	0.9672	0.9699	0.9684	0.9661	0.9699	0.7501	0.7325
0.4	0.0286	0.9640	0.9669	0.9653	0.9628	0.9669	0.7180	0.6966
0.5	0.0308	0.9620	0.9652	0.9635	0.9605	0.9652	0.6998	0.6772

Table 1: Performance comparison of models with different training datasets.

3D Logits Distribution Across Different Mask Probabilities

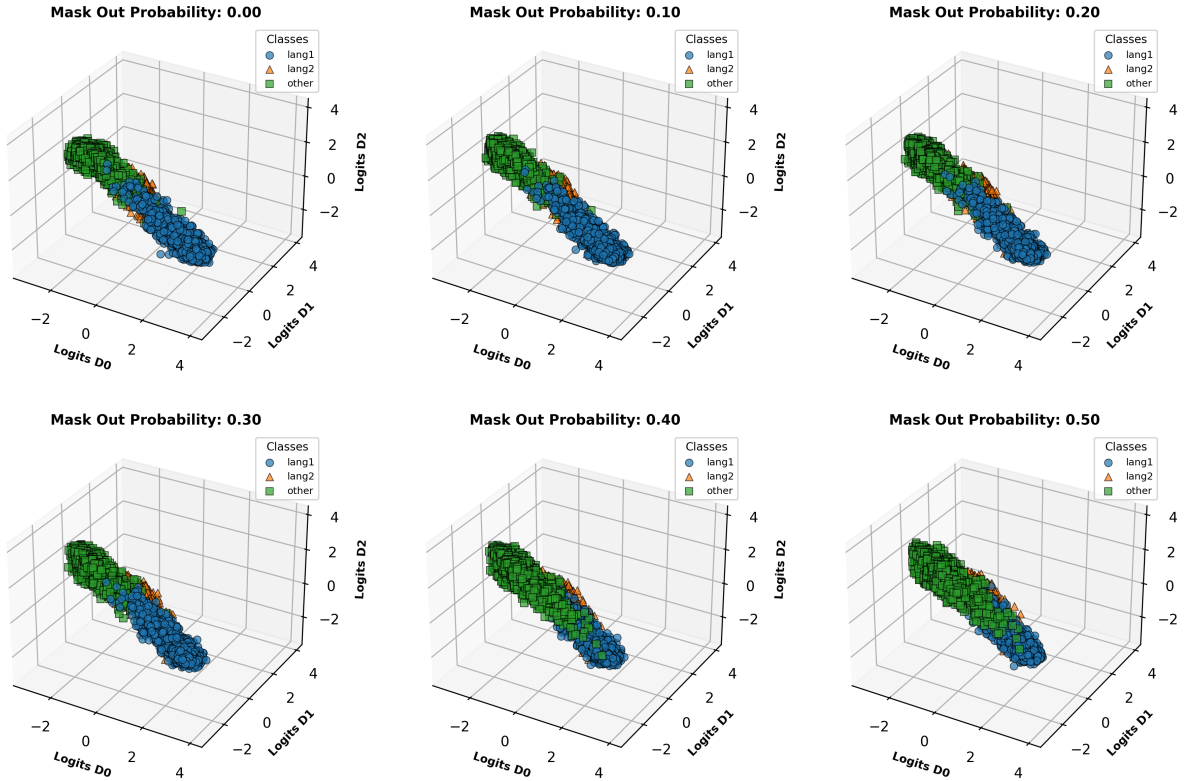


Figure 5: 3D visualization of logits distribution.

including named entity recognition, sentiment analysis, and sequence labeling in machine translation, by improving model performance and generalization. Future research could investigate this hypothesis by applying the method to these tasks and evaluating its effectiveness in diverse linguistic and code-switching contexts.

Additionally, further exploration could focus on optimizing mask-out probabilities to better suit specific tasks or datasets, as well as examining alternative regularization techniques that complement token masking. These efforts could provide deeper insights into the underlying mechanisms of regularization and contribute to the development of more

robust models for multilingual and code-switching tasks.

8 Limitations

Despite the promising results of this study, there are several limitations to our approach:

- Dataset Generalization:** The model is trained and evaluated on SPA-ENG and NEP-ENG datasets from the LinCE benchmark. While the results demonstrate generalization across these datasets, the findings may not extend to other language pairs or less-studied code-switching contexts, particularly those involving non-Indo-European languages or languages with substantial structural

2D Logits Distribution: Dim 1 vs Dim 2 Across Different Mask Probabilities

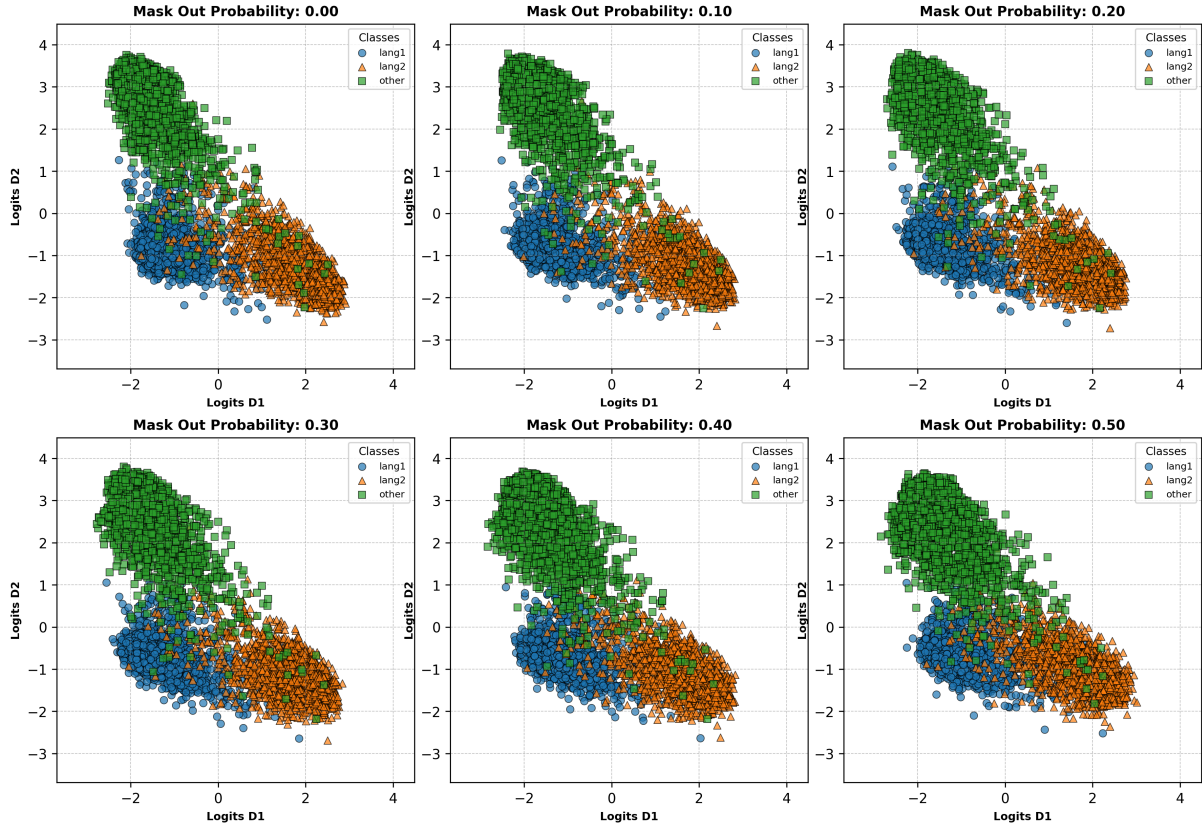


Figure 6: 2D visualization of logits distribution.

differences.

2. Mask-Out Probability Tuning: While a mask-out probability of 0.1 yielded optimal results in this study, this value may not generalize to other datasets or tasks. The process of tuning this hyperparameter could be computationally intensive, especially for larger datasets or more complex code-switching tasks.

3. Class Imbalance: Although class weighting is employed to address label imbalance, this technique may not fully mitigate the challenges posed by highly imbalanced datasets. This limitation could affect the model’s robustness and predictive accuracy for minority classes.

4. Lack of Interpretability: Transformer-based models like mBERT are inherently opaque, making it difficult to fully understand the decision-making process. This limitation restricts our ability to verify that the model is leveraging meaningful linguistic patterns rather than spurious correlations.

9 Ethical Concerns

The following ethical considerations arise from this study:

1. Bias Propagation: Pretrained models like mBERT may encode biases present in their training data, such as stereotypes or culturally inappropriate representations. These biases could propagate into downstream tasks, leading to unfair or inaccurate predictions, particularly for certain languages, dialects, or user groups.

2. Resource Requirements: The computational demands of fine-tuning mBERT and training large-scale models contribute to the carbon footprint of AI research. This raises concerns about the environmental sustainability of large-scale NLP projects.

3. Ethical Data Use: Although the LinCE datasets are well-established benchmarks, the use of social media data requires strict adherence to data usage agreements and ethical guidelines. Given the personal and potentially sensitive nature of this data, it is essential to ensure that data privacy and user consent are upheld.

10 Group Member Task Breakdown

Xianglong Xu: Developed the majority of the code base, including the implementation of the proposed token-masking regularization method and training scripts, and played a key role in drafting the final report.

John Bowen: Contributed by revising and refining portions of the code to enhance functionality and ensure robustness, and prepared the project presentation.

Rojin Taheri: Generated the visualizations and figures to illustrate performance metrics and dataset distributions, conducted the analysis of results and wrote significant portions of the report.

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